

Electronic Components and Circuit Design

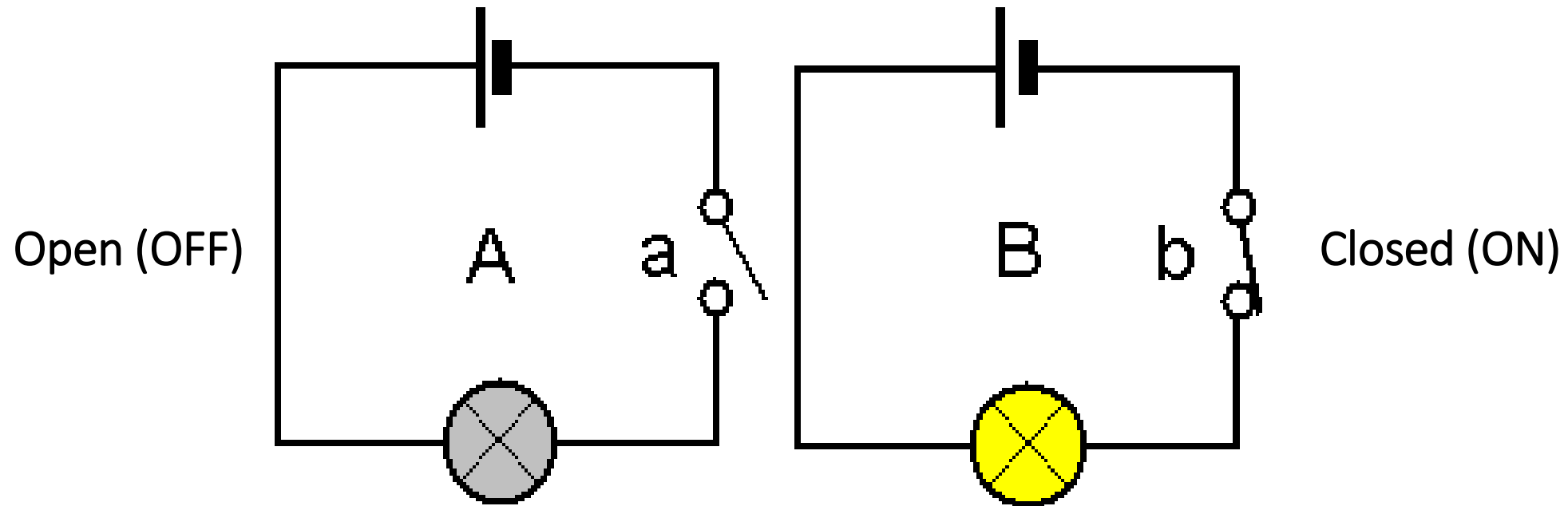
Electronics & Physical Computing



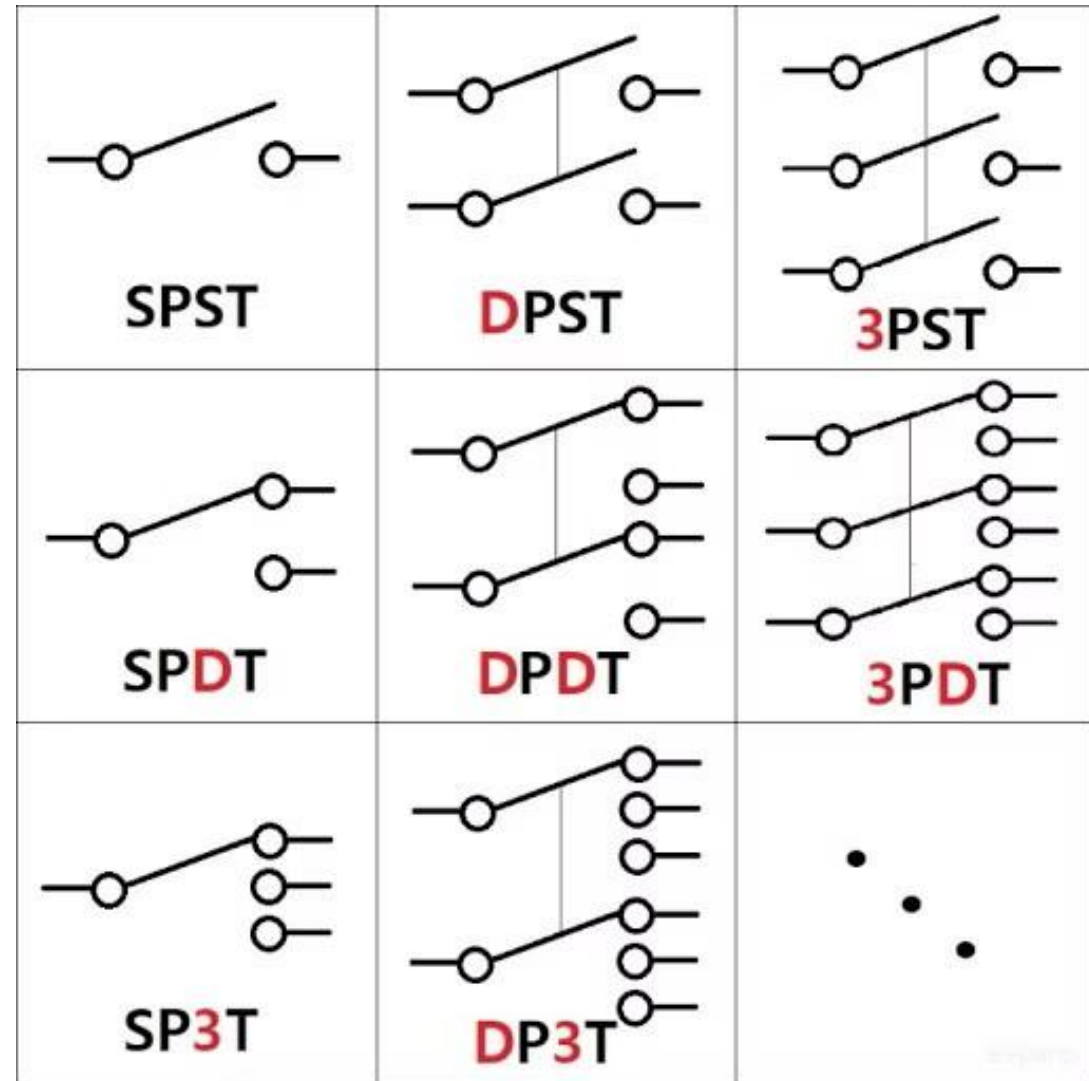
Switches

What is a Switch?

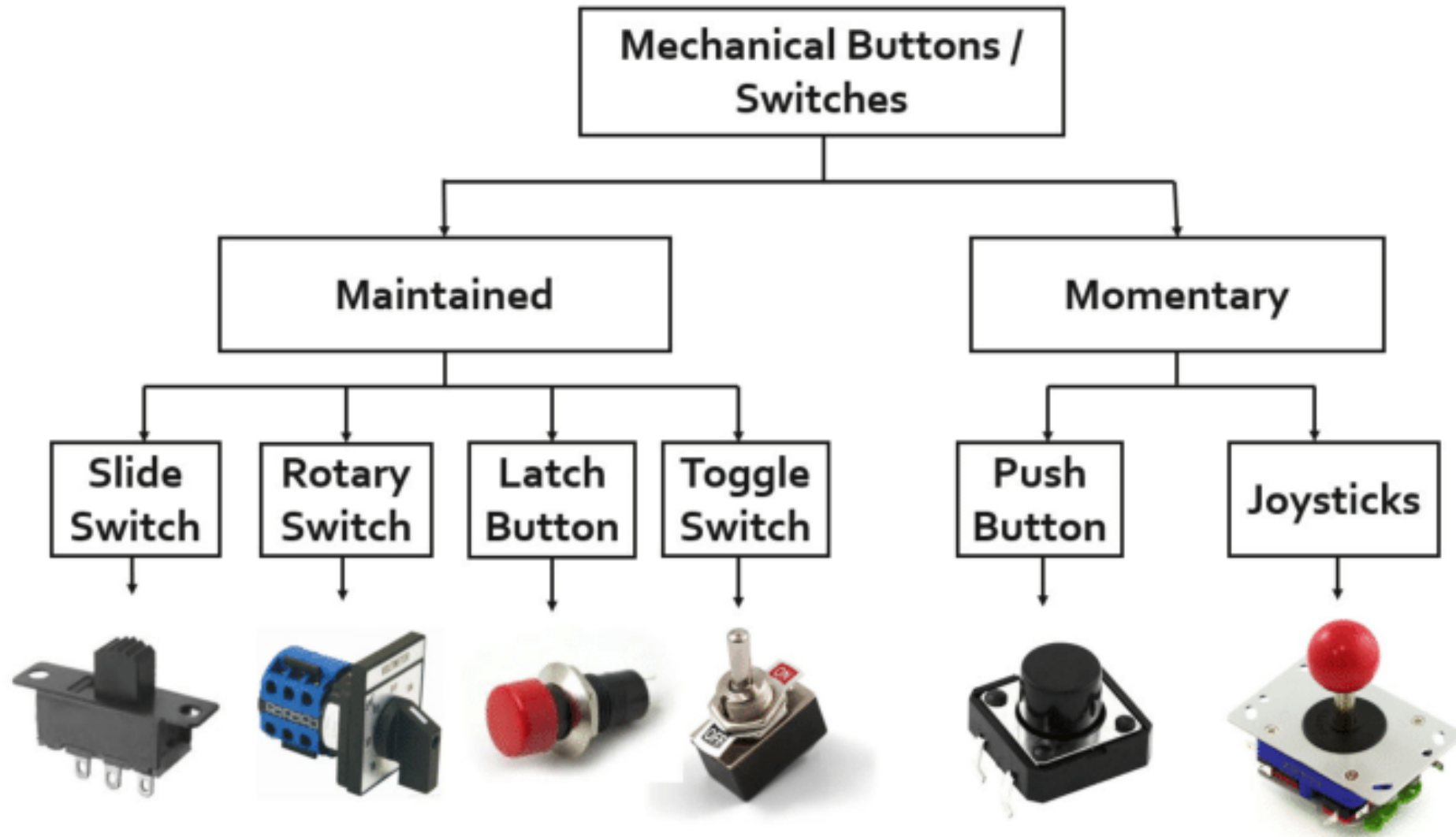
A passive electromechanical device used to open or close an electrical circuit, thereby starting or stopping the flow of current.



Types of Switches #1 (Based on Pole & Throw)



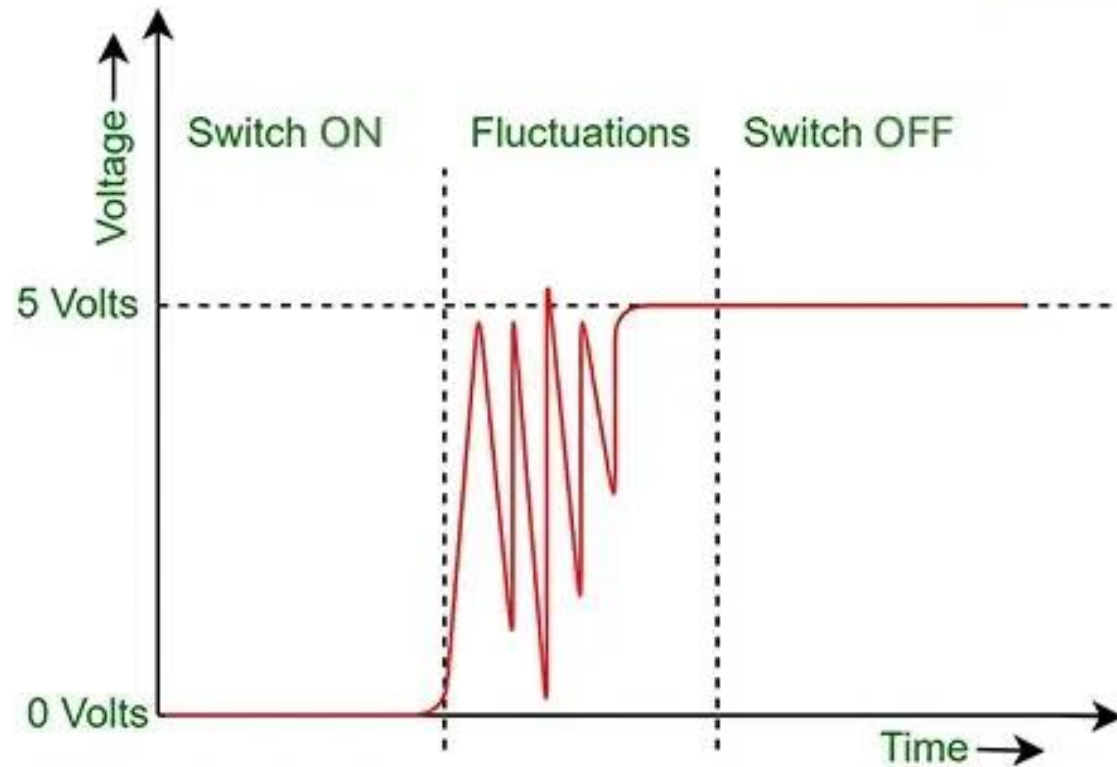
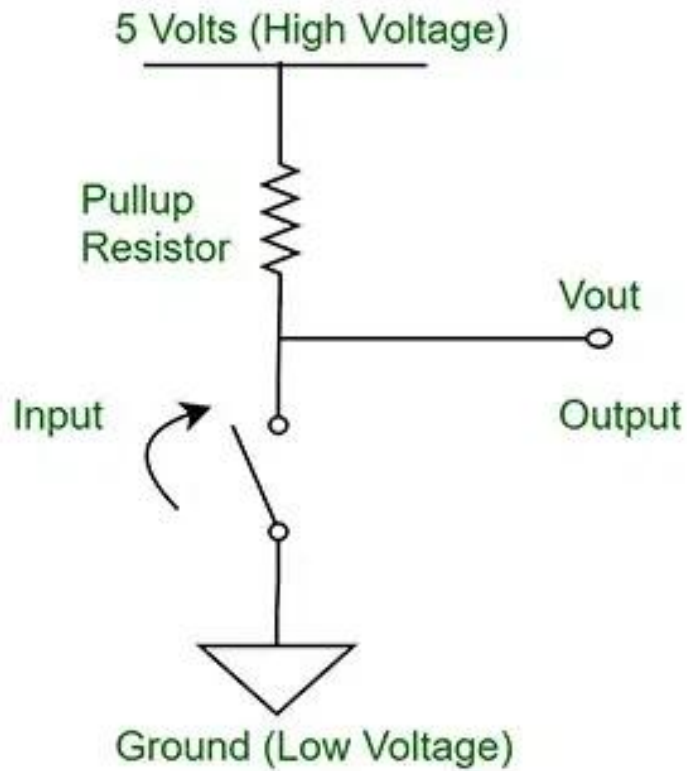
Types of Switches #2 (Based on Operation)



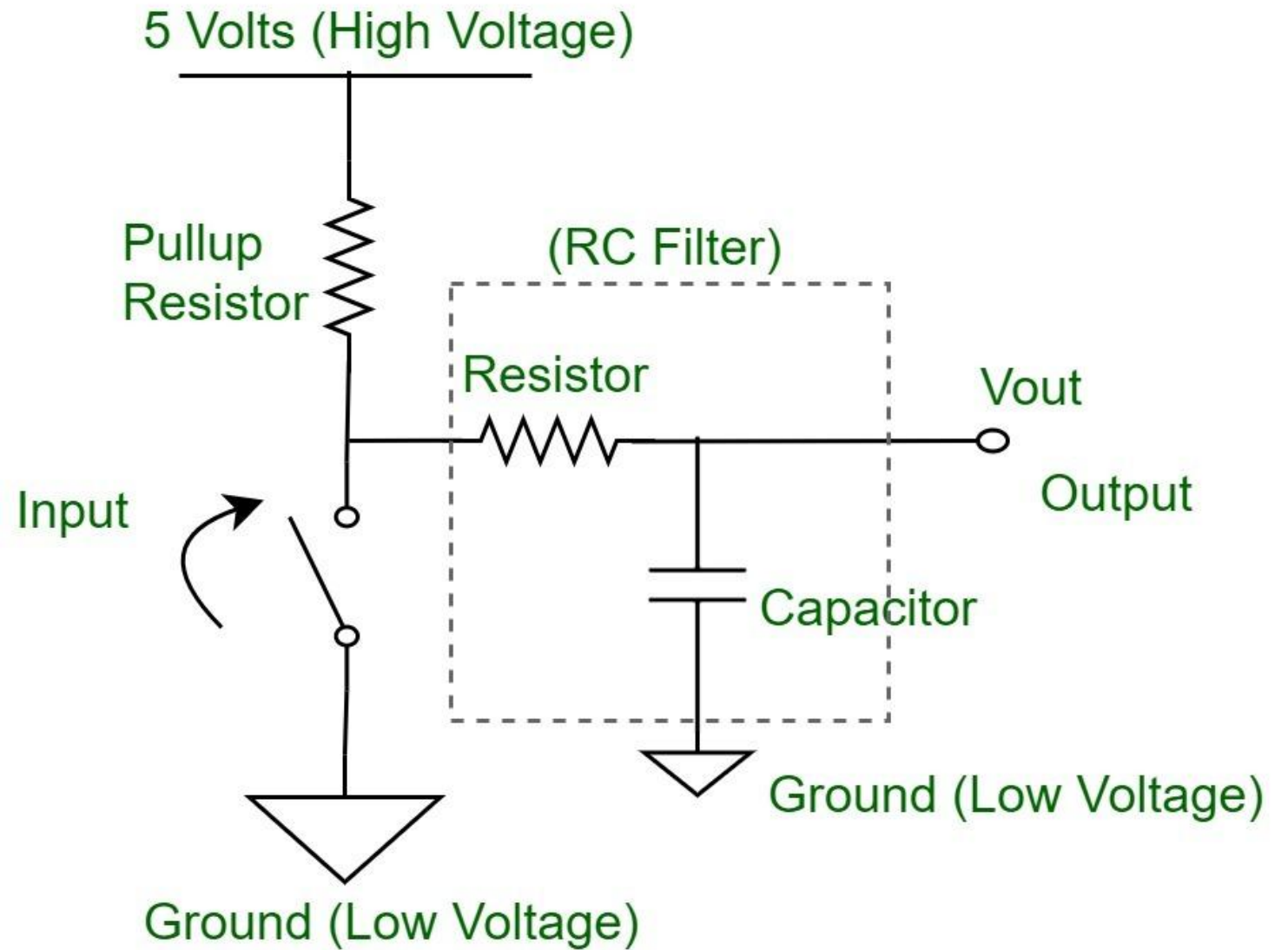


Switch Ratings

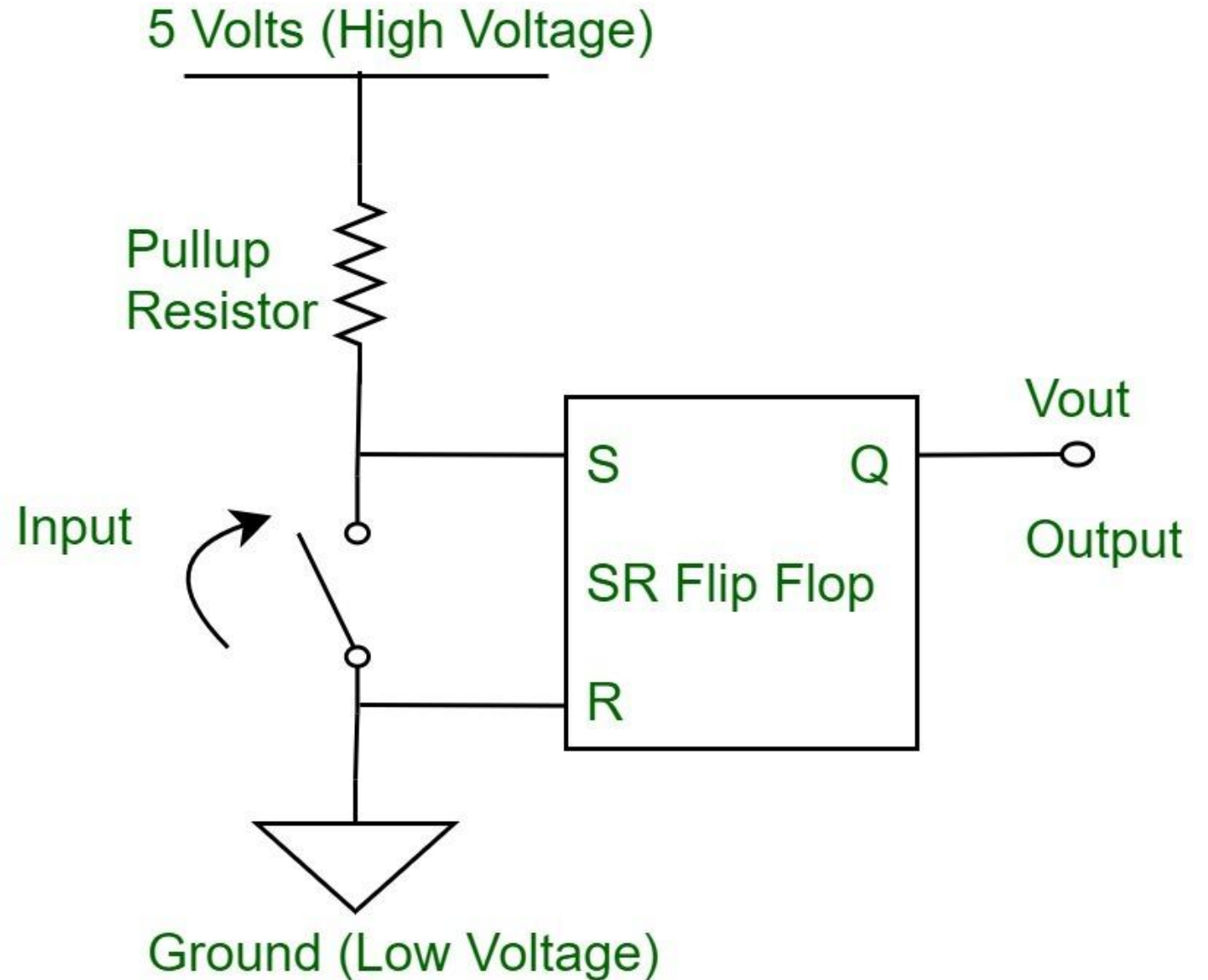
Switch Bouncing

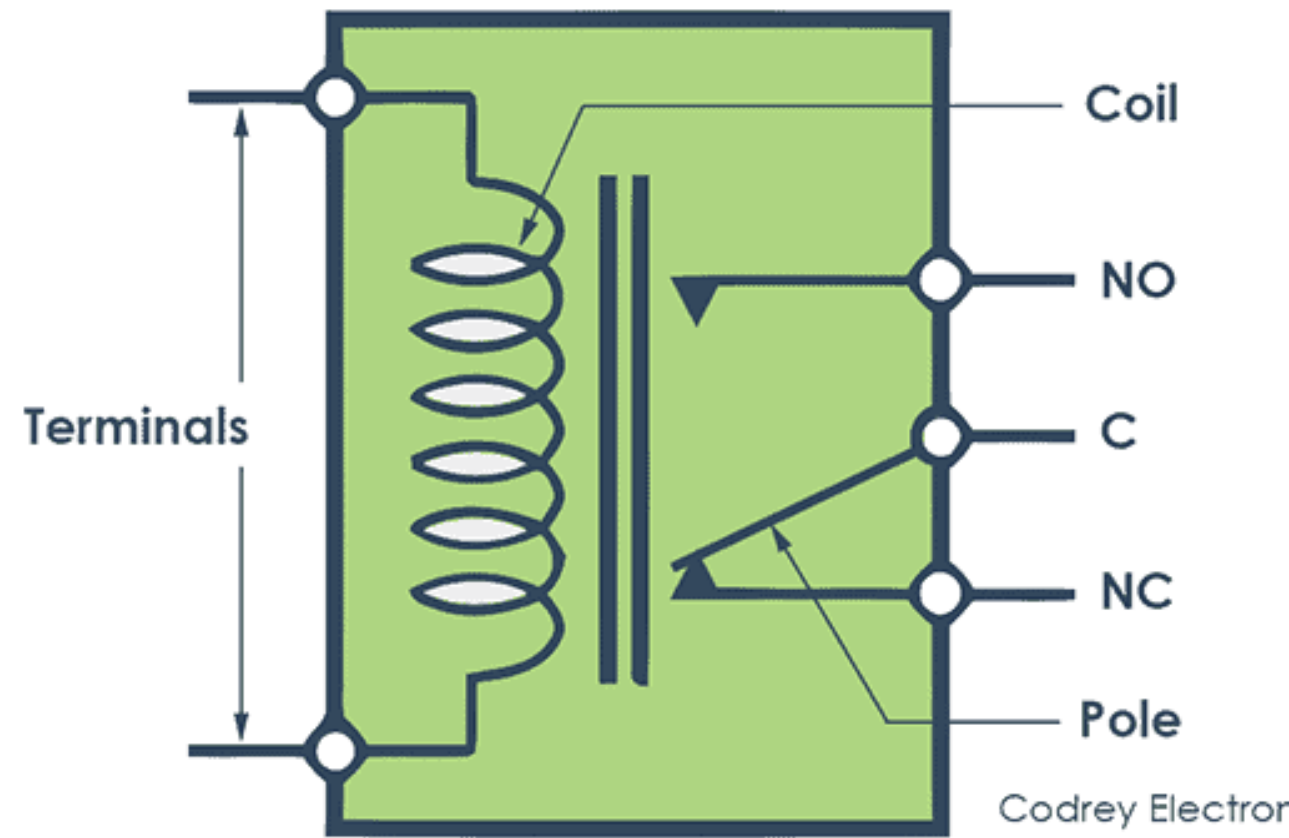


Switch Debouncing Solutions #1



Switch Debouncing Solutions #2



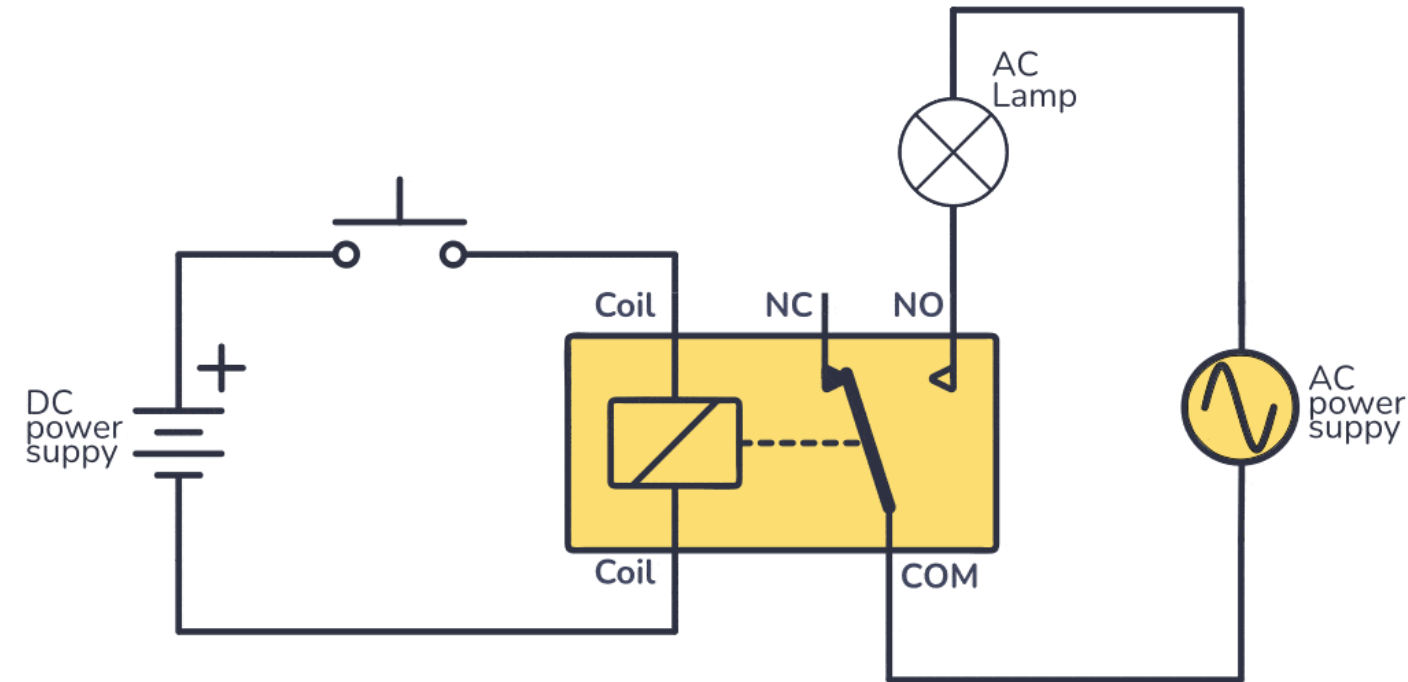


Relays

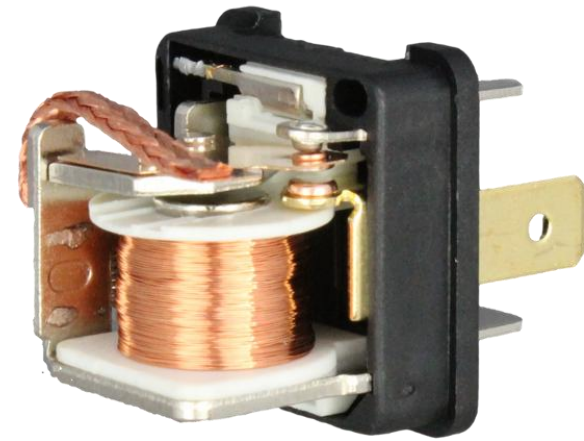
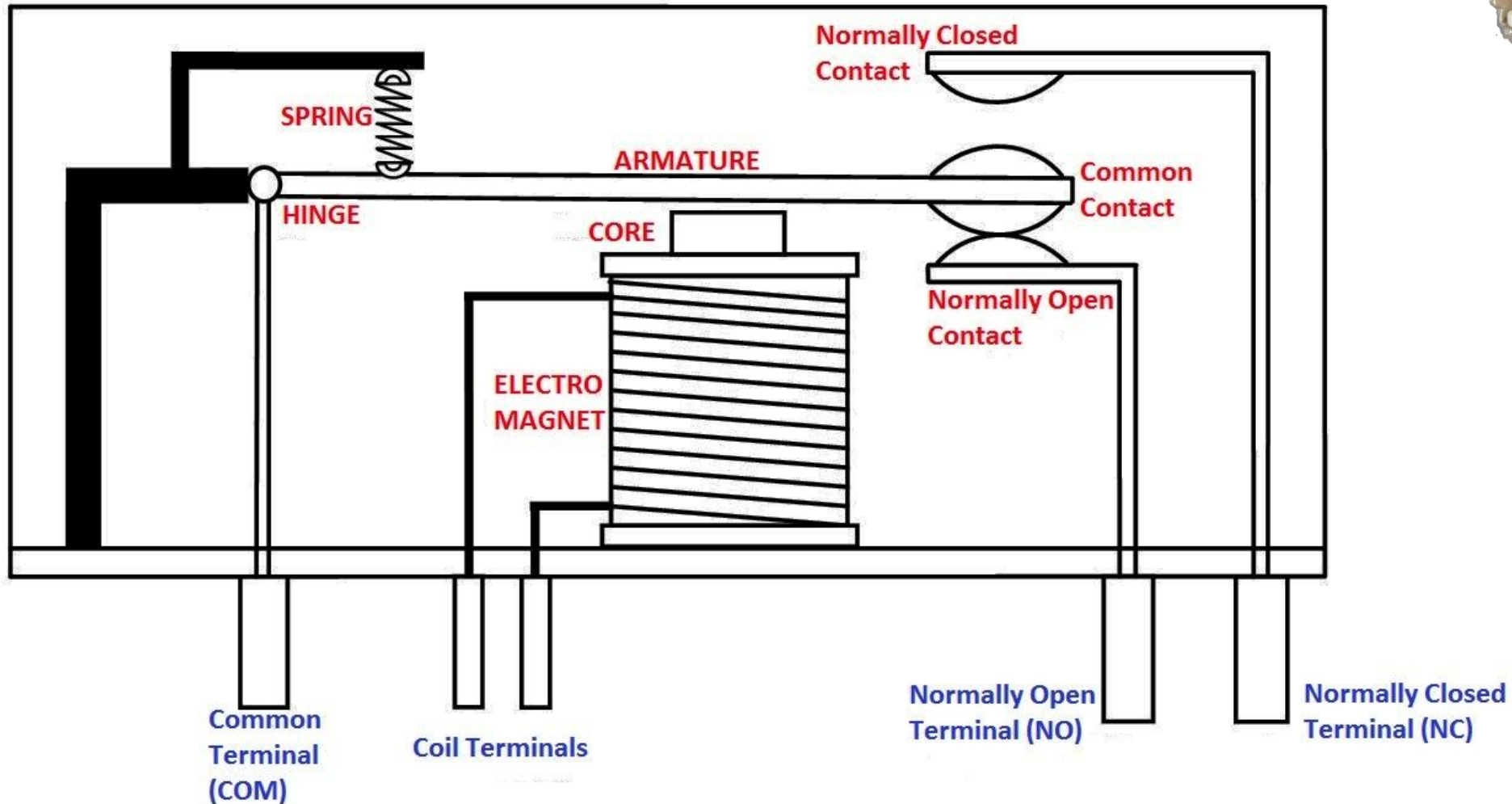
What is a Relay?

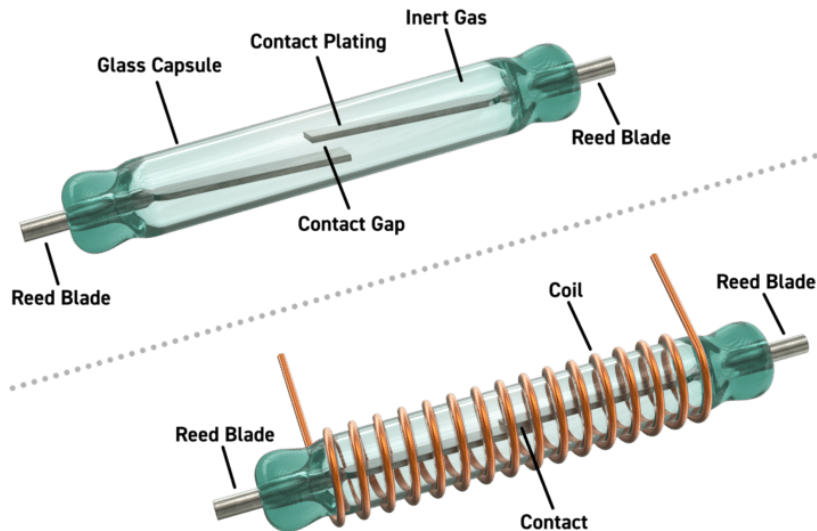
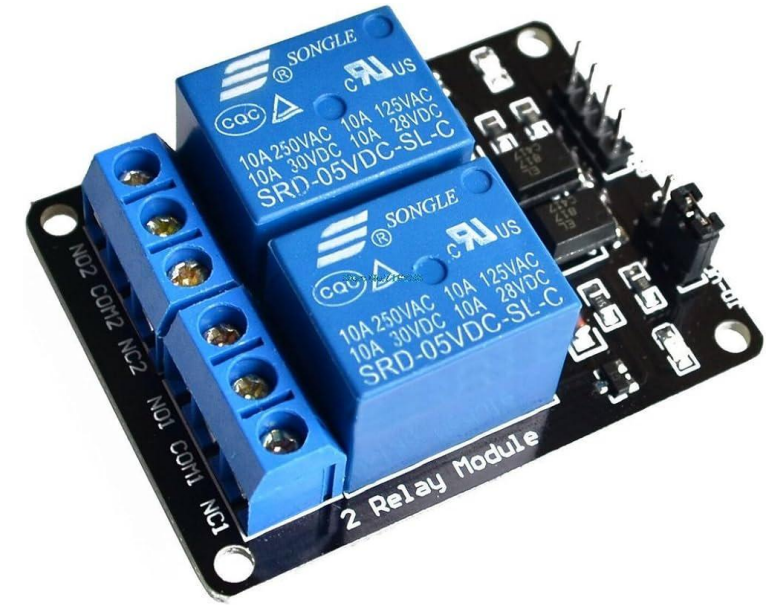
An **electromechanical switch** that uses an **electromagnet** to open or close one or more contacts.

It allows a **low-power signal** (like from an Arduino) to **control a high-power** circuit (like a motor, bulb, or AC appliance) together with **electrical isolation**.

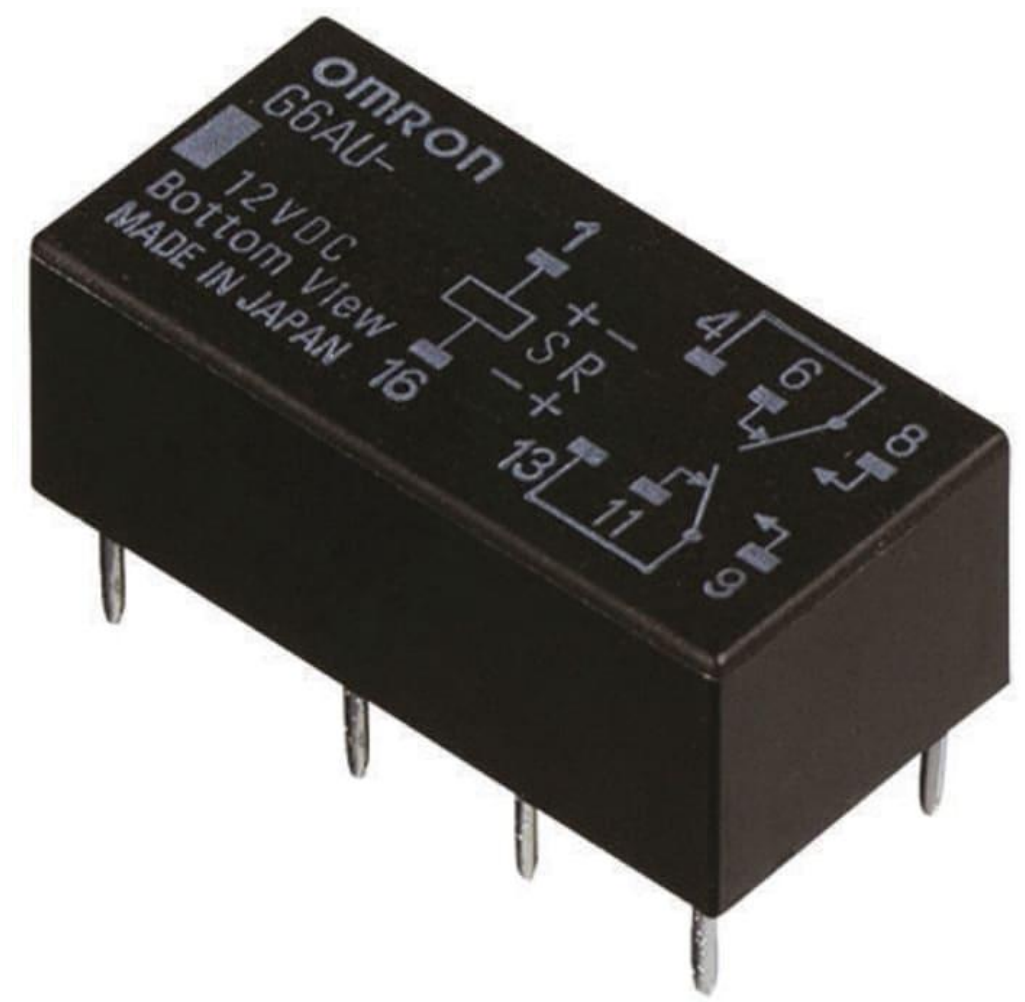
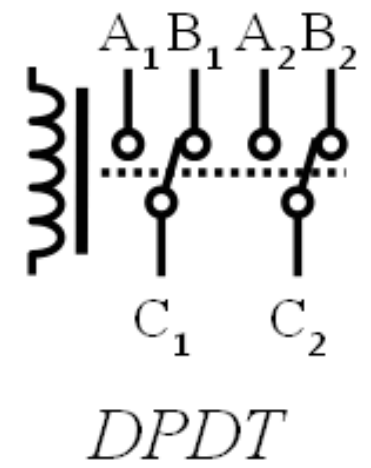
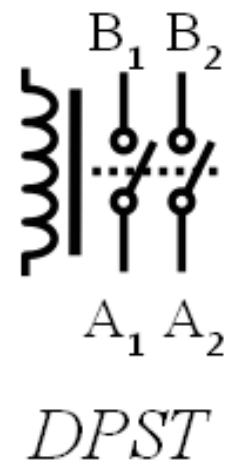
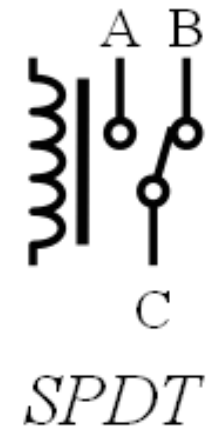


How Does a Relay Work?



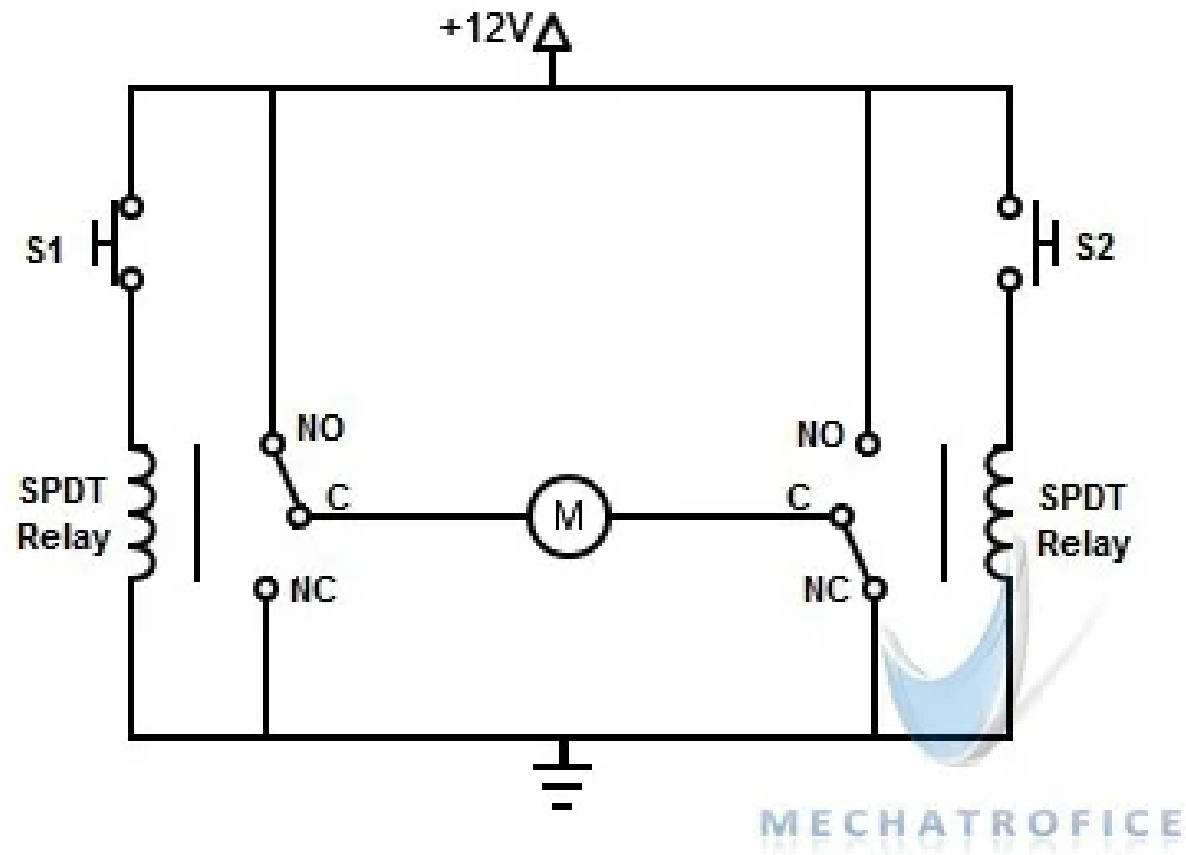


Types of Relays



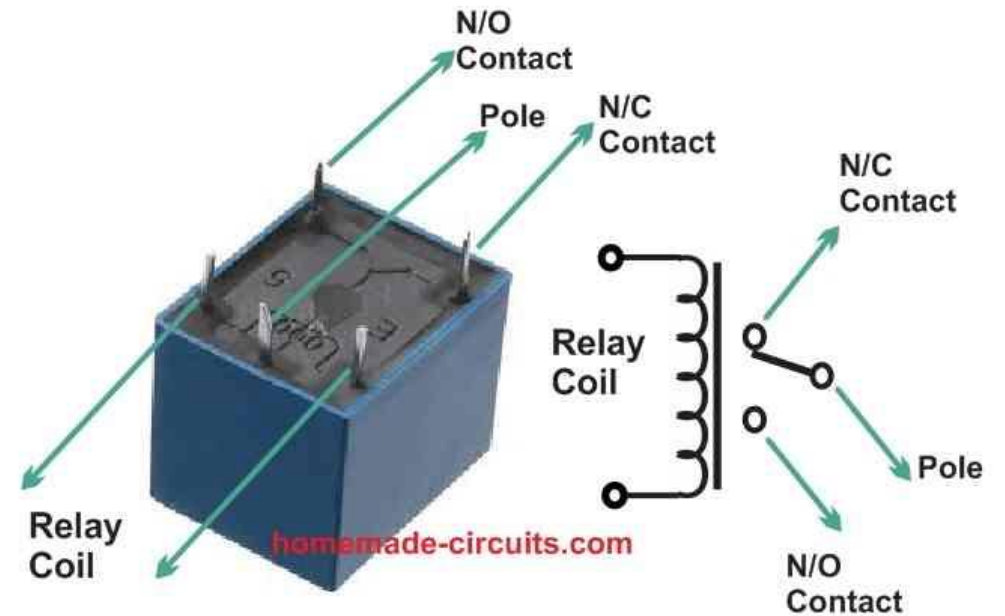
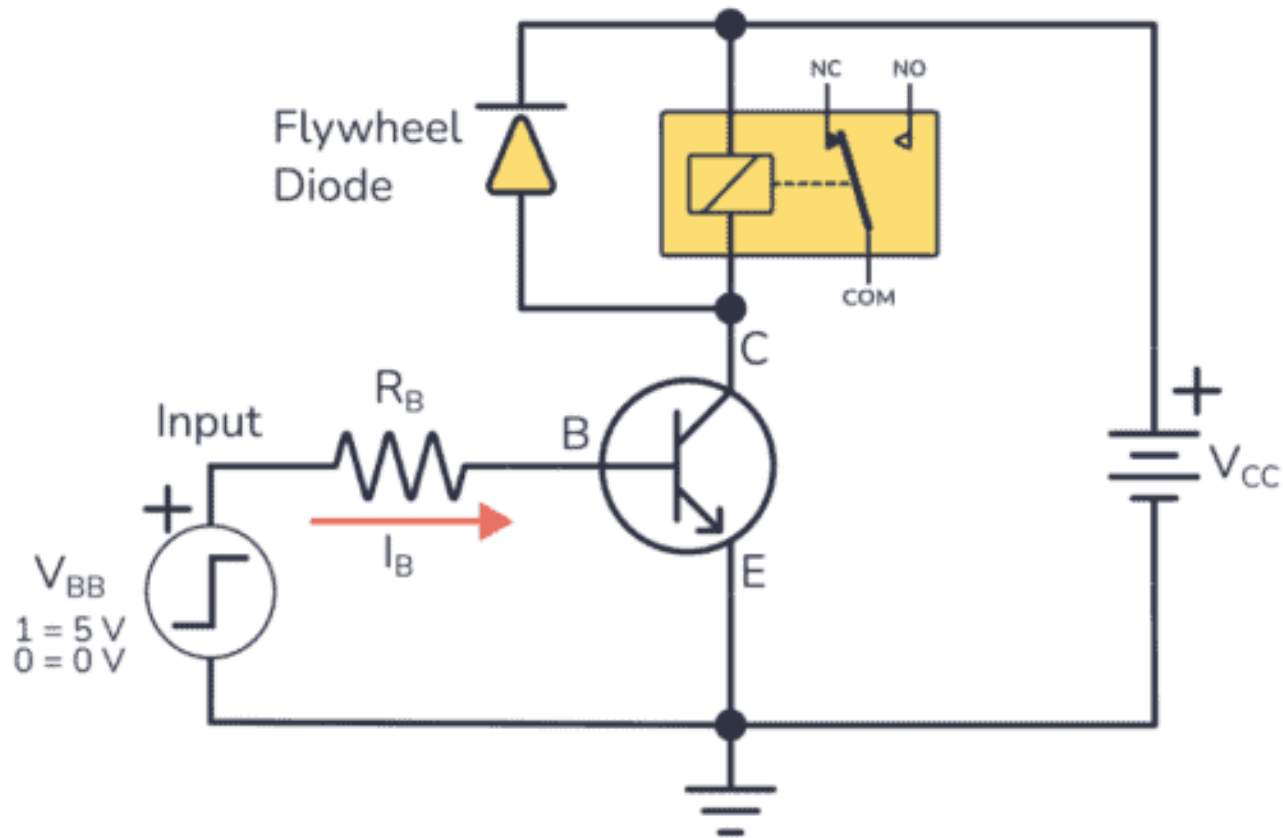


Relay Parameters/Ratings



Applications of Relays

Connecting Relays



N/C = Normally Closed
N/O = Normally Open



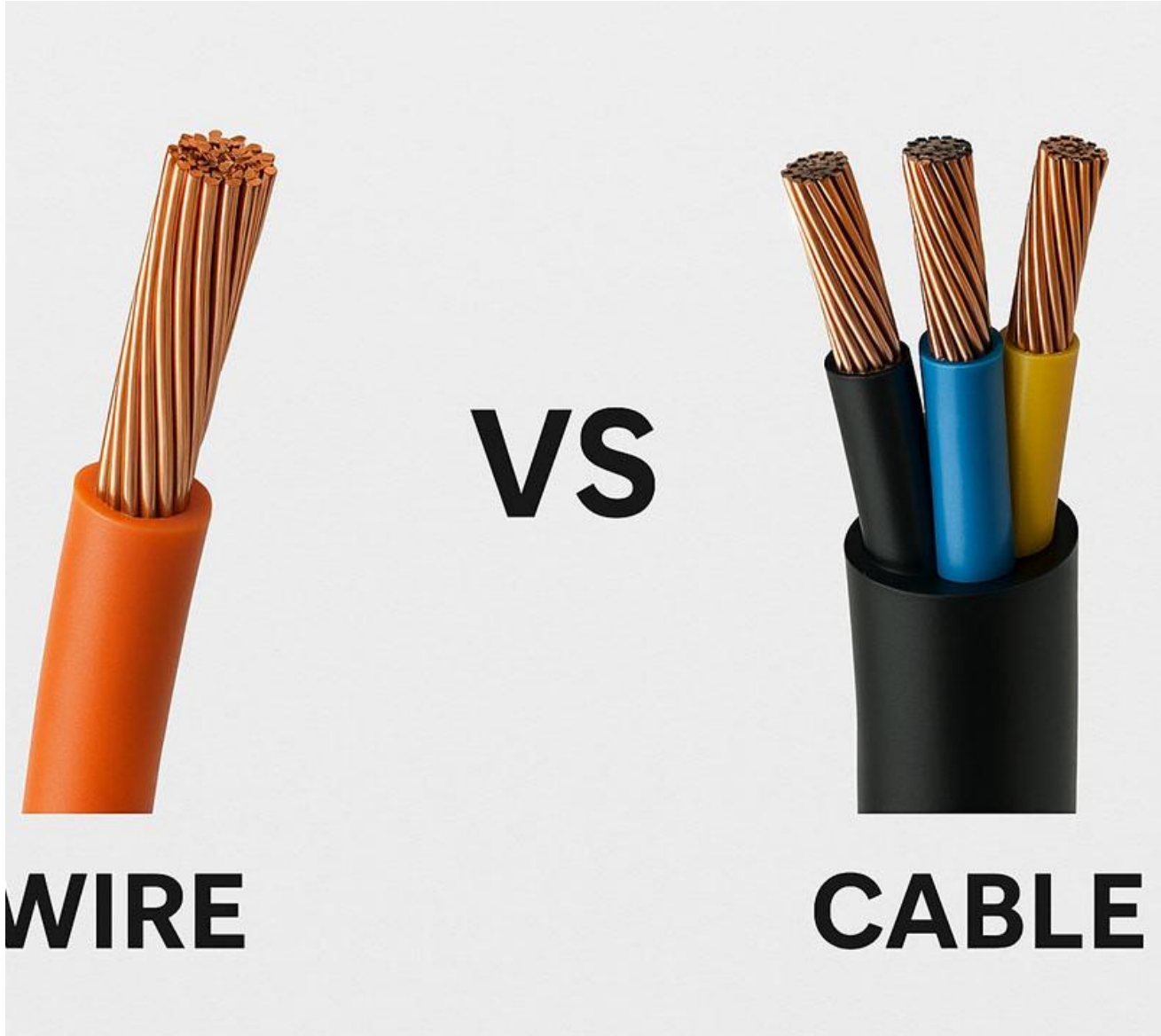
VS



WIRE

CABLE

Wires & Cables



What Are Wires and Cables?

Wire: a single conductor

Cable: a bundle of wires

Stranded vs. Solid Wires

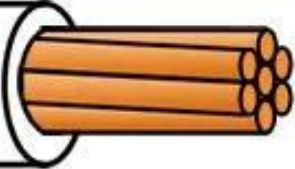
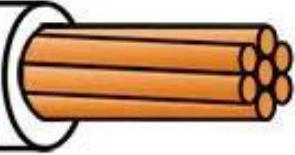


Stranded



Solid

Wire Ampacity Chart

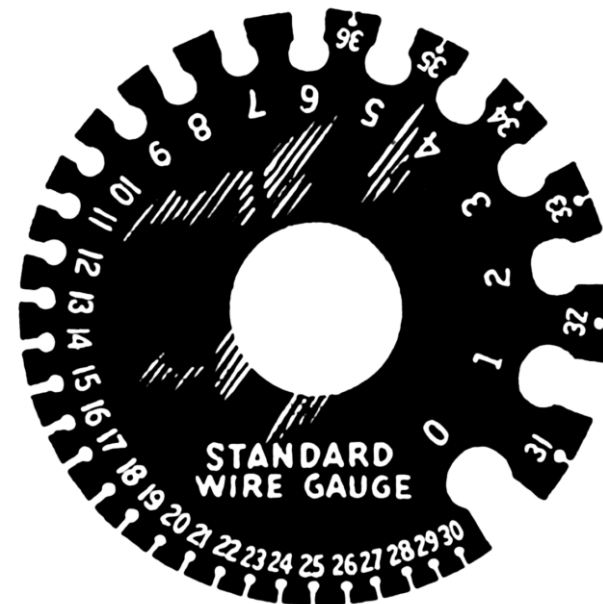
#3 Gauge		100 amps
#6 Gauge		65 amps
#8 Gauge		50 amps
#10 Gauge		30 amps
#12 Gauge		20 amps
#14 Gauge		15 amps

Copper wire for 75°C THW, THWN, SE type wires © FAQware

Wire Gauge (Thickness)

Measured using AWG (American Wire Gauge) or mm^2 .

Lower AWG => Thicker Wire => More Current





Pure Copper Wire



2.5mm	(7/0.67)	Earth Wire
1.5mm	(7/0.53)	Single Wire
1.5mm	(7/0.53)	Twin Wire
2.5mm	(7/0.67)	Single Wire
2.5mm	(7/0.67)	Twin Wire
6mm	(7/1.04)	Single Wire

Commercial Wires

Calculation



1mm	(1/1.13)	Single Wire
2.5mm	(7/0.67)	Earth Wire
1.5mm	(7/0.53)	Single Wire
1.5mm	(7/0.53)	Twin Wire
2.5mm	(7/0.67)	Single Wire
2.5mm	(7/0.67)	Twin Wire
6mm	(7/1.04)	Single Wire
6mm	(7/1.04)	Twin Wire
0.5mm	(16/0.20)	2Core Wire
0.5mm	(16/0.20)	T/T Wire

Meaning of 7/0.53:

- **7** = Number of strands in the wire (i.e., it is a stranded wire with 7 individual copper strands).
- **0.53 mm** = Diameter of each strand.

Calculating total cross-sectional area:

Each strand has a diameter of 0.53 mm, so:

$$\text{Area of one strand} = \frac{\pi}{4} \times (0.53)^2 \approx 0.2206 \text{ mm}^2$$

$$\text{Total area} = 7 \times 0.2206 \approx 1.544 \text{ mm}^2$$

So, a 7/0.53 wire has a total cross-sectional area of about 1.5 mm^2 , which is a standard wire size used in electrical installations.

Rated for around **10–13 amps**, depending on insulation and installation conditions.

Colour Coding (Common Practice)

Color	Typical Meaning (DC)
Red	Positive voltage (+)
Black	Ground (GND) or negative
Yellow	Signal or 3.3V
Blue	Signal or data
Green	Signal or Earth Ground

Function	India Color Code (Old)	India Color Code (New)
Single Phase Line		
Single Phase Neutral		
Single Phase Protective Ground or Earth		
Three Phase Line (L1)		
Three Phase Line (L2)		
Three Phase Line (L3)		
Three Phase Neutral (N)		
Three Phase Protective Earth or Ground (PE)		

 Color codes can vary by country and standard — always double-check when working with AC or mains wiring.

Connectors

